



HK IMO Training 2021-2022
Problem Set 25
For 9 April 2022



Problem 97. Define a sequence $\{a_n\}$ by $a_1 = -3$ and $a_{n+1} = a_n + 4 + 4\sqrt{a_n + 4}$. Find a_{2022} .

Problem 98. Let $\triangle ABC$ be an acute triangle with $\angle A = 45^\circ$. Let AD, BE, CF be the altitudes of $\triangle ABC$, and let H be the orthocentre of $\triangle ABC$. Let P be the point on the side AB such that $AP = CH$. Suppose $BD : DC = 2 : 3$. Prove that H is the centroid of $\triangle BCP$.

Problem 99. Find all positive integers m and n such that each of them divides $9^{|m-n|} + 3^{|m-n|} + 1$.

Problem 100. There are 2022 magnetic bars. One end of each magnetic bar is red in colour, while the other end is blue in colour. For some magnetic bars of type 1, the red end has north magnetic polarity and the blue end has south magnetic polarity. For the other magnetic bars of type 2, the red end has south magnetic polarity and the blue end has north magnetic polarity. Each time we can put two magnetic bars together to test whether they are of the same type or of different types. We want to know whether there are equally many type 1 magnetic bars as type 2 magnetic bars. At least how many comparisons are needed so that we are guaranteed to know the answer?